Lab Assignment 4 – OpenPGP

The Lab consisted of generating an OpenPGP key pair using Kleopatra from the Gpg4win encryption suite (Outlook on Windows). During the key generation process, a passphrase was assigned to protect the private key from unauthorized use. After creation, the key pair was saved in the local keyring and the associated Key ID and Fingerprint were recorded for future verification and communication.

Next, the teaching assistant's public key (b544ta@gmail.com) was retrieved from a public OpenPGP key server. The fingerprint of the downloaded key was inspected and verified before being imported into Kleopatra to establish a trusted communication channel. Following this, the GpgOL plugin which is included in the Gpg4win installation was enabled in Microsoft Outlook to allow encryption and digital signing of outgoing messages.

After the configuration, the student sent three required test emails to the TA:

1. A digitally signed email
2. A signed and encrypted email
3. An encrypted-only email

For each message, the recipient's public key was used to encrypt the content, while the sender's private key was used to generate the digital signature when applicable. These operations were automatically handled by Outlook through the GpgOL interface. Confirmation messages and status indicators verified successful encryption and signature generation before transmission.

To enable secure responses from others, the generated OpenPGP public key was also published to a public key server (keys.openpgp.org) This step ensures that others can securely obtain the public key to initiate encrypted communication.

OpenPGP Key ID: 9A9C BC6F E5C0 B288
Key Fingerprint: 5CA1419218348101EB2179E39A9CBC6FE5C0B288

Q. Can you send signed/encrypted email using webmail such as Gmail? Why or why not? What are the challenges?

Sending signed or encrypted email directly through standard Gmail webmail is not supported because Gmail does not provide built-in OpenPGP functionality. It does not have access to a user's private key, which is required for signing and decrypting messages. Although it is possible with third-party browser plugins, this approach introduces challenges such as key management, lack of native integration, and weaker trust verification compared to desktop clients like Outlook or Thunderbird.

Q. Given the fingerprint of the key, why are you convinced that the key file you downloaded must not have been tampered with?

The key is verified by comparing its unique fingerprint with the expected fingerprint provided by the owner, since fingerprints are generated by a cryptographic hash, a match ensures the key has not been tampered with or replaced.